

Determinants of Insurance Premiums: Lifestyle Choices vs. Demographic Factors

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1. Introduction

Health insurance premiums in the United States are determined by complex actuarial algorithms. For the consumer, the weighting of these factors is often hard to interpret. While some determinants of cost are **Fixed Demographic Factors** (Age, Region, Sex), others are **Modifiable Lifestyle Choices** (Smoking, BMI, Alcohol Consumption).

Understanding the distinction between these categories is vital. If premiums are driven primarily by demographics, consumers have little impact over their costs. However, if lifestyle factors are significant drivers, individuals can mitigate their financial burden through behavioral changes.

Research Question: **To what extent do modifiable lifestyle habits (Smoking, BMI, Alcohol) explain the variance in insurance premiums compared to fixed demographic factors?**

The Data: We utilize the Medical Insurance Cost Prediction dataset from Kaggle [1]. The dataset contains approximately 100,000 observations including demographic details (Age, Sex, Income) and lifestyle indicators (Smoking status, BMI, Alcohol frequency).

2. Exploratory Data Analysis

2.1 Summary Statistics

Table 1 summarizes the central tendencies of the continuous variables. The substantial difference between the median (463.58 USD) and mean (582.32 US\$) premium indicates a right-skewed distribution, suggesting the presence of high-cost outliers.

Table 1: Summary of Numerical Variables

Average Age	Average BMI	Median Premium	Mean Premium	Max Premium
47.52	26.99	463.58	582.32	10962.55

2.2 Distribution and Transformation

Linear regression relies on the assumption of normality in the error terms. As shown in Figure 1 (Left), the raw annual premium distribution is heavily right-skewed. To satisfy the normality assumption, we applied a log-transformation ($\ln(Premium)$). Figure 1 (Right) confirms that the transformed variable approximates a normal distribution, validating the use of log-linear regression [2].

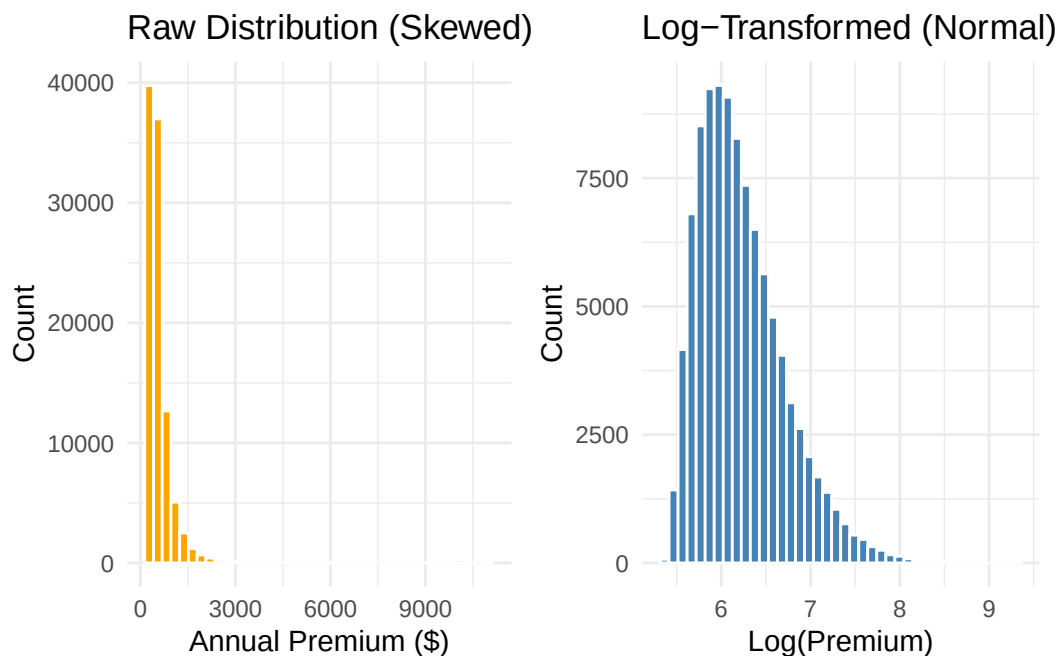


Figure 1

2.3 Visualizing the “Smoking Penalty” and Interaction

We hypothesized that the cost burden of smoking might vary by age. Figure 2 plots Age against Log-Premium, stratified by smoking status.

The Green Line (Current Smokers) shows a higher intercept, indicating a higher baseline cost. More importantly, the slope appears steeper than that of the Red Line (Never Smokers). This suggests a potential interaction effect: the financial penalty for smoking compounds as the individual ages. We will test this formally in our regression models.

2.4 BMI Trends

We also examined the relationship between BMI and cost. Figure 3 shows a positive (though weak) linear relationship between BMI and the log of insurance premiums. This still allows to consider BMI as a relevant predictor to include in the “Modifiable Lifestyle” category.

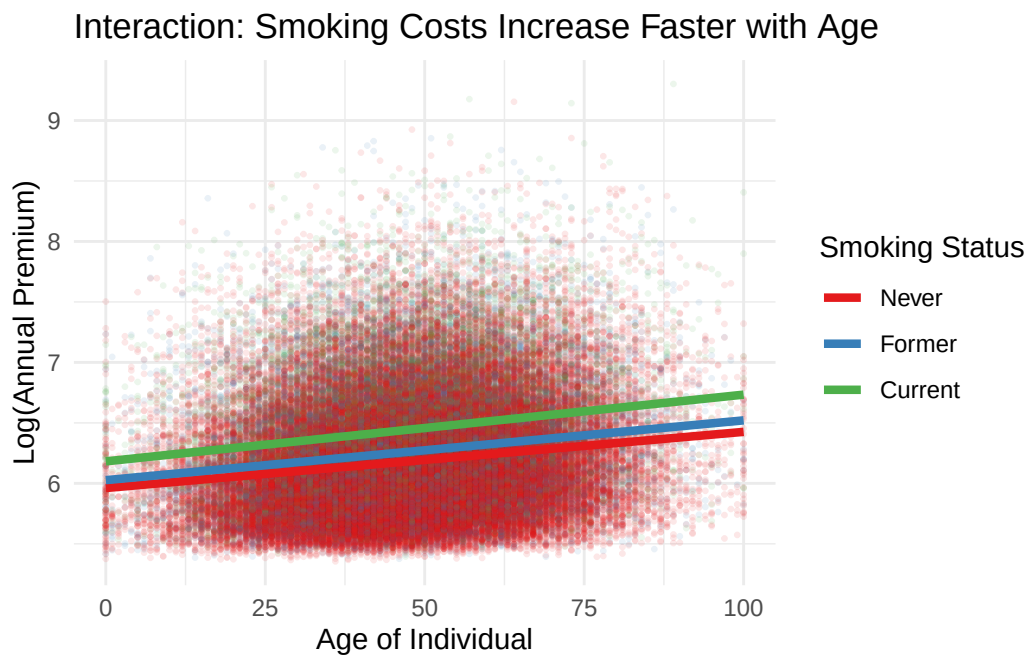


Figure 2

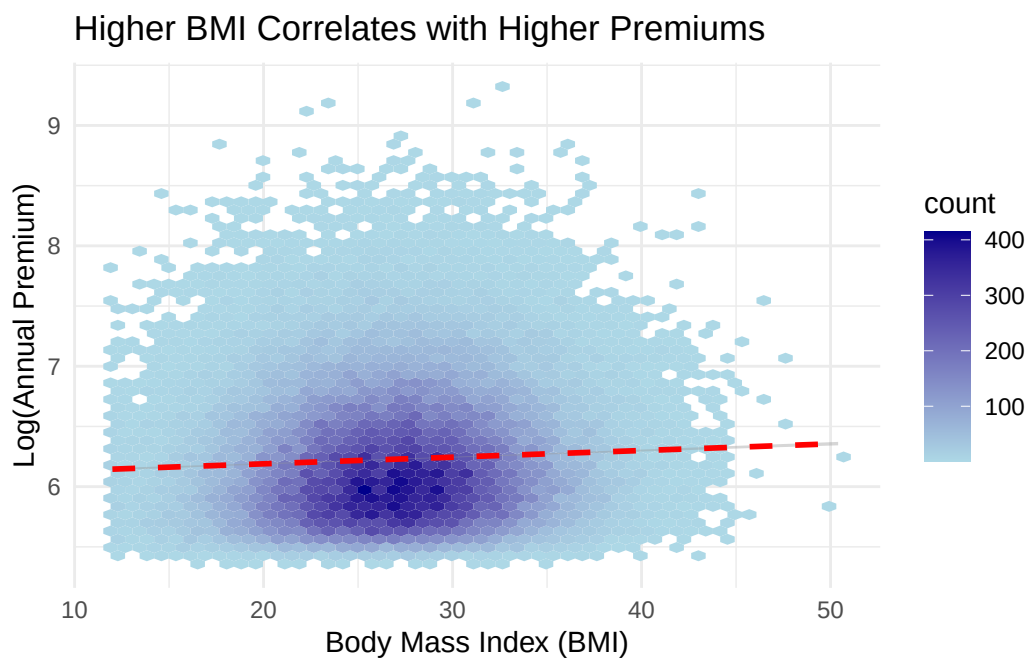


Figure 3

3. Methodology

To isolate the variance explained by lifestyle choices, we employ Hierarchical Multiple Linear Regression[3]. We fit two distinct models:

Model 1: Baseline Demographics This model accounts for fixed factors the individual cannot easily change.

$$\ln(Y) = \beta_0 + \beta_{Age} + \beta_{Sex} + \beta_{Region} + \beta_{Income} + \dots + \epsilon$$

Model 2: Full Model (Demographics + Lifestyle) This model adds modifiable behaviors. Based on the interaction observed in Section 2.3, we include an interaction term for Age and Smoking.

$$\ln(Y) = \text{Model 1} + \beta_{BMI} + \beta_{Smoker} + \beta_{Alcohol} + (\beta_{Age} \times \beta_{Smoker}) + \epsilon$$

We utilize Analysis of Variance (ANOVA) to statistically compare the two models. The F-test in the ANOVA will determine if the reduction in the Residual Sum of Squares (RSS) provided by the lifestyle variables is statistically significant.

4. Results

4.1 ANOVA Model Comparison

We performed an ANOVA F-test to compare the explanatory power of Model 1 (Demographics) versus Model 2 (Full).

Table 2: ANOVA Comparison: Baseline vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99980	22900.88	NA	NA	NA	NA
99972	22085.80	8	815.09	461.19	0

Interpretation: The ANOVA table shows a highly significant result for Model 2

- F-Statistic: The large F-value (461.19) implies that the variance explained by the added lifestyle variables is significantly greater than what we would expect by random chance
- P-Value (< 0.001): Since $p < 0.05$, we reject the null hypothesis. We can conclude that adding Smoking, BMI, and Alcohol significantly improves the model fit over demographics alone. Note: The NA in the first row is normal, as it represents the baseline model against which the second model is compared.

4.2 Regression Coefficients

Table 3 presents the coefficients for the Full Model. Since the response is log-transformed, coefficients (β) can be interpreted as an approximate $100 \times \beta\%$ change in the premium for small values.

Table 3: Coefficients for Modifiable Lifestyle Factors (Model 2)

Predictor	Estimate	Std. Error	P-Value
BMI	0.00562	0.0002976	<0.001
Smoker: Former	0.06373	0.0123031	<0.001
Smoker: Current	0.22166	0.0143493	<0.001
alcohol_freqOccasional	-0.00062	0.0034994	0.859
Alcohol: Weekly	-0.00388	0.0042996	0.367
Alcohol: Daily	0.01245	0.0071756	0.083
age:Smoker: Former	0.00032	0.0002462	0.193
age:Smoker: Current	0.00087	0.0002870	0.002

Key Findings:

1. Smoking: Being a “Current Smoker” has a large positive coefficient (0.221), indicating a substantial premium increase compared to never-smokers, holding other factors constant.
2. BMI: The coefficient for BMI is positive and statistically significant ($p < 0.001$). Higher body mass index is reliably associated with higher insurance costs.
3. Alcohol: The p-values for alcohol frequency are > 0.05 . This suggests that, unlike smoking and BMI, self-reported alcohol frequency is not a statistically significant predictor of insurance premiums in this dataset.
4. Interaction (Age $\times$ Smoker): The significant interaction term ($p = 0.002$) confirms our EDA finding. The slope of the cost curve is steeper for smokers. This means the dollar-cost of smoking increases as the individual ages.

4.3 Diagnostics and Assumptions Checking

To ensure the validity of our inference, we examined the residual plots for Model 2.

- Linearity and Homoscedasticity: Figure 4 (Left) displays the Residuals vs. Fitted values. The points are randomly distributed around the horizontal zero line without a distinct funnel shape. This indicates that the assumption of homoscedasticity (constant variance of errors) is largely met.
- Normality of Residuals: Figure 4 (Right) shows the Normal Q-Q plot. The points (circles) adhere closely to the diagonal line for the majority of observations. However, the tails deviate upward. This “heavy-tailed” behavior indicates that our model slightly underestimates the premiums for the most extreme, high-cost outliers. Given the large sample size ($N \approx 100,000$), the Central Limit Theorem ensures that our coefficient estimates remain unbiased despite this minor deviation.

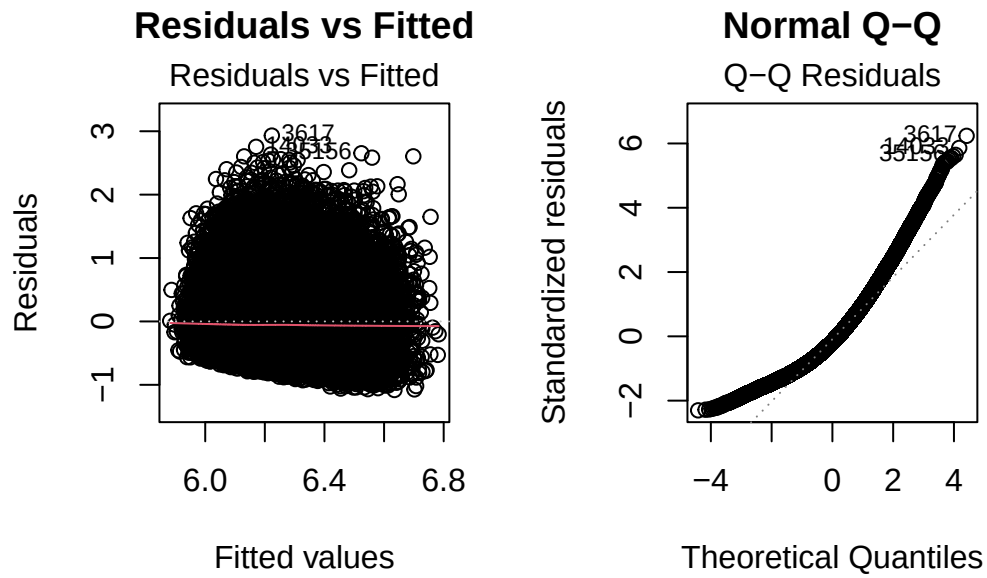


Figure 4

5. Discussion

6. References

[1] Kaggle. (2024). Medical Insurance Cost Prediction Dataset. [2] Wooldridge, J. M. (2015). Introductory Econometrics. [3] UVa Library. (2023). Hierarchical Linear Regression. <https://library.virginia.edu/data/articles/hierarchical-linear-regression>